

Chap 7. Operations on controlled rough paths

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May 1st, 2020

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1. Relation between $\mathcal{C}^\alpha$ and $\mathcal{D}_X^{2\alpha}$
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6. Summary

Relation between $\mathcal{C}^\alpha$ and $\mathcal{D}_X^{2\alpha}$

Theorem (4.10. (Gubinelli))

Let $T > 0$, let $\mathbf{X} = (X, \mathbb{X}) \in \mathcal{C}^\alpha([0, T], V)$ for some $\alpha > 1/3$, and let $(Y, Y') \in \mathcal{D}_X^{2\alpha}([0, T], \mathcal{L}(V, W))$. Then there is a constant C depending only on α such that for every pair s, t

$$\left| \int_s^t Y_r d\mathbf{X}_r - Y_s X_{s,t} - Y'_s \mathbb{X}_{s,t} \right| \leq C(\|X\|_\alpha R^Y_{2\alpha} + \|\mathbb{X}\|_{2\alpha} Y'_\alpha) |t - s|^{3\alpha}$$

if the integral of Y against $\mathbf{X}$ exists.

Relation between $\mathcal{C}^\alpha$ and $\mathcal{D}_X^{2\alpha}$

Remark 4.11. Consider $(Z, Z') \in \mathcal{D}_X^{2\alpha}([0, T], \bar{V})$ with above assumption.

We may define

$$\int_s^t Y_u dZ_u \stackrel{\text{def}}{=} (\mathcal{I}\Xi)_{s,t}, \quad \Xi_{u,v} = Y_u Z_{u,v} + Y'_u Z'_u \mathbb{X}_{u,v}.$$

Here, $Z'_u \in \mathcal{L}(V, \bar{V}) \cong \mathcal{L}(V \otimes V, V \otimes \bar{V})$, and

$Y'_u \in \mathcal{L}(V, \mathcal{L}(\bar{V}, W)) \cong \mathcal{L}(V \otimes \bar{V}, W)$

Also the above inequality is valid,

$$\begin{aligned} & \left| \int_s^t Y_r dZ_r - Y_s Z_{s,t} - Y'_s Z'_s \mathbb{X}_{s,t} \right| \\ & \leq C(\|X\|_\alpha \|R^Z\|_{2\alpha} + \|\mathbb{X}\|_{2\alpha} \|Y'Z'\|_\alpha) |t - s|^{3\alpha} \end{aligned}$$

Relation between $\mathcal{C}^\alpha$ and $\mathcal{D}_X^{2\alpha}$

Relation between $\mathcal{C}^\alpha$ and $\mathcal{D}_X^{2\alpha}$.

$$\mathbf{X} = (X, \mathbb{X}) \in \mathcal{C}^\alpha([0, T], V)$$

1. X can be interpreted as a path controlled by X .
 X can be identified with $(X, I) \in \mathcal{D}_X^{2\alpha}$
2. $(Y, Y') \in \mathcal{D}_X^{2\alpha}([0, T], W)$ can be interpreted as a rough path, say $\mathbf{Y} = (Y, \mathbb{Y})$.

$$\mathbb{Y}_{s,t} = \int_s^t Y_{s,r} \otimes dY_r \stackrel{\text{def}}{=} \lim_{|\mathcal{P}| \rightarrow 0} \sum_{\mathcal{P}} \Xi,$$
$$\Xi_{u,v} = Y_u \otimes Y_{u,v} + Y'_u \otimes Y'_v \mathbb{X}_{u,v}.$$

Thus there is a continuous canonical injection

$$\mathcal{D}_X^{2\alpha}([0, T], W) \hookrightarrow \mathcal{C}^\alpha([0, T], W)$$

Relation between $\mathcal{C}^\alpha$ and $\mathcal{D}_X^{2\alpha}$

Exercise 7.7. Verify that $\mathbb{X}_{s,t} = \int_s^t X_{s,r} \otimes dX_r$ where the integral is to be interpreted in the sense of Remark 4.11., taking (Y, Y') to be (X, I)
Proof.

$$\begin{aligned} & \sum_{[u,v] \in \mathcal{P}} X_u \otimes X_{u,v} + I \otimes I \mathbb{X}_{u,v} \\ &= \sum_{[u,v] \in \mathcal{P}} X_u \otimes X_{u,v} + \mathbb{X}_{u,v} \end{aligned}$$

Denote $X_u \otimes X_{u,v} + \mathbb{X}_{u,v}$ by $\Xi_{u,v}$. Recall that the Chen's relation is given by

$$\mathbb{X}_{u,w} - \mathbb{X}_{u,v} - \mathbb{X}_{v,w} = X_{u,v} \otimes X_{v,w}.$$

By using this relation, we have that for any w

$$\Xi_{u,v} + \Xi_{v,w} = \Xi_{u,w}$$

By taking $w = v$, we may complete the proof. □

This interpretation of elements of $\mathcal{D}_X^{2\alpha}$ as elements of $\mathcal{C}^\alpha$ is coherent in terms of the theory of integration constructed in the previous section, as can be seen by the following result:

Proposition (Proposition 7.1. in Textbook)

Let $(X, \mathbb{X}) \in \mathcal{C}^\alpha$, let $(Y, Y') \in \mathcal{D}_X^{2\alpha}$, and let $\mathbf{Y} = (Y, \mathbb{Y}) \in \mathcal{C}^\alpha$ be the associated rough path constructed as above. If $(\tilde{Z}, \tilde{Z}') \in \mathcal{D}_Y^{2\alpha}$, then $(Z, Z') \in \mathcal{D}_X^{2\alpha}$, where $Z_t = \tilde{Z}_t$ and $Z'_t = \tilde{Z}'_t Y'_t$. Furthermore

$$\int_0^t Z_s dY_s = \int_0^t \tilde{Z}_s d\mathbf{Y}_s. \quad (1)$$

Relation between $\mathcal{C}^\alpha$ and $\mathcal{D}_X^{2\alpha}$

Proof.

1. By assumption,

$$Y_{s,t} = Y'_s X_{s,t} + O(|t-s|^{2\alpha})$$

$$\tilde{Z}_{s,t} = \tilde{Z}'_s Y_{s,t} + O(|t-s|^{2\alpha})$$

$$\Rightarrow Z_{s,t} = \tilde{Z}'_s Y'_s X_{s,t} + O(|t-s|^{2\alpha}) = Z'_s X_{s,t} + O(|t-s|^{2\alpha})$$

$$\Rightarrow (Z, Z') \in \mathcal{D}_X^{2\alpha}.$$

2. Let $\Xi_{s,t} = Z_s Y_{s,t} + Z'_s Y'_s \mathbb{X}_{s,t}$ and $\tilde{\Xi}_{s,t} = \tilde{Z}_s Y_{s,t} + \tilde{Z}'_s \mathbb{Y}_{s,t}$

$$(LHS \text{ of (1)}) = \mathcal{I}\Xi_{0,t}$$

$$(RHS \text{ of (1)}) = \mathcal{I}\tilde{\Xi}_{0,t}$$

Since $|\mathbb{Y}_{s,t} - Y'_s Y'_s \mathbb{X}_{s,t}| \leq C|t-s|^{3\alpha}$ by Theorem 4.10, the claim follows from Remark 4.11 □

Remark.

If $\frac{1}{3} < \beta < \alpha$, then $\mathcal{C}^\alpha \hookrightarrow \mathcal{C}^\beta$.

For every $X \in \mathcal{C}^\alpha$, there is a canonical embedding $\mathcal{D}_X^{2\alpha} \hookrightarrow \mathcal{D}_X^{2\beta}$

Lifting of regular paths

There is a canonical embedding $\iota : \mathcal{C}^{2\alpha} \hookrightarrow \mathcal{D}_X^{2\alpha}$ given by $\iota Y = (Y, 0)$, since $R_{s,t} = Y_{s,t}$ does satisfy $\|R\|_{2\alpha} < \infty$.

Recall that we are only interested in the case $\alpha \leq 1/2$. After all, if $Y_{s,t} = O(|t-s|^{2\alpha})$ with $\alpha > 1/2$, then Y has a vanishing derivative and must be constant.

Chain rule for the Gubinelli derivative

Chain rule for the Gubinelli derivative

Lemma (7.3. in Textbook)

Let $\varphi \in \mathcal{C}_b^2$, $(Y, Y') \in \mathcal{D}_X^{2\alpha}([0, T], W)$ for some $X \in \mathcal{C}^\alpha$ with $|Y'_0| + \|Y, Y'\|_{X, 2\alpha} \leq M \in [1, \infty)$. Let $(\varphi(Y), \varphi(Y)') \in \mathcal{D}_X^{2\alpha}([0, T], \bar{W})$ be given by

$$\varphi(Y)_t = \varphi(Y_t), \quad \varphi(Y)'_t = D\varphi(Y_t)Y'_t.$$

Then, there exist a constant C depending only on $T > 0$ and $\alpha > \frac{1}{3}$ such that one has the bound

$$\|\varphi(Y), \varphi(Y)'\|_{X, 2\alpha} \leq C_{\alpha, T} M \|\varphi\|_{\mathcal{C}_b^2} (1 + \|X\|_\alpha)^2 (|Y'_0| + \|Y, Y'\|_{X, 2\alpha})$$

At last, C can be chosen uniformly over $T \in (0, 1]$

Chain rule for the Gubinelli derivative

Proof.

We have $(\varphi(Y), \varphi(Y)') = (\varphi(Y.), D\varphi(Y.)Y.') \in \mathcal{D}_X^{2\alpha}$.

$$\begin{aligned}\|\varphi(Y)\|_\alpha &\leq \|D\varphi\|_\infty \|Y.\|_\alpha \\ \|\varphi(Y)'\|_\alpha &\leq \|D\varphi(Y.)\|_\infty \|Y.'\|_\alpha + \|Y.'\|_\infty \|D\varphi(Y.)\|_\alpha \\ &\leq \|D\varphi(Y.)\|_\infty \|Y.'\|_\alpha + \|Y.'\|_\infty \|D^2\varphi(Y.)\|_\infty \|Y.\|_\alpha\end{aligned}$$

which shows that $\varphi(Y), \varphi(Y)' \in \mathcal{C}^\alpha$. And $R^\varphi \equiv R^{\varphi(Y)}$ is given by

$$\begin{aligned}R_{s,t}^\varphi &= \varphi(Y_t) - \varphi(Y_s) - D\varphi(Y_s)Y_s'X_{s,t} \\ &= \varphi(Y_t) - \varphi(Y_s) - D\varphi(Y_s)Y_{s,t} + D\varphi(Y_s)R_{s,t}^Y\end{aligned}$$

so that

$$\|R^\phi\|_{2\alpha} \leq \frac{1}{2} |D^2\phi|_\infty \|Y\|_\alpha^2 + |D\phi|_\infty \|R^Y\|_{2\alpha}.$$

Chain rule for the Gubinelli derivative

It follows that

$$\begin{aligned}\|\varphi(Y), \varphi(Y)'\|_{X, 2\alpha} &\leq \|D\varphi(Y.)\|_{\infty} \|Y.\|_{\alpha} + \|Y.\|_{\infty} \|D^2\varphi(Y.)\|_{\infty} \|Y.\|_{\alpha} \\ &\quad + \frac{1}{2} |D^2\varphi|_{\infty} \|Y\|_{\alpha}^2 + |D\varphi|_{\infty} \|R^Y\|_{2\alpha} \\ &\leq \|\varphi\|_{\mathcal{C}_b^2} (\|Y.\|_{\alpha} + \|Y.\|_{\infty} \|Y.\|_{\alpha} + \|Y\|_{\alpha}^2 + \|R^Y\|_{2\alpha}) \\ &\leq C_{\alpha, \tau} \|\varphi\|_{\mathcal{C}_b^2} (1 + \|X\|_{\alpha})^2 (1 + |Y'_0| + \|Y, Y'\|_{X, 2\alpha}) \\ &\quad \times (|Y'_0| + \|Y, Y'\|_{X, 2\alpha}).\end{aligned}$$

The last inequality is followed from the def. of $\|Y, Y'\|_{X, 2\alpha}$ and the following inequality (which is in the textbook (4.18)). □

$$\|Y\|_{\alpha} \leq C(1 + \|X\|_{\alpha})(|Y'_0| + \|Y, Y'\|_{X, 2\alpha})$$

Corollary (7.4. in Textbook)

Let (Y, Y') and (Z, Z') be two controlled paths in $\mathcal{D}_X^{2\alpha}$ for some $X \in \mathcal{C}$. Then the path $U = YZ$, with Gubinelli derivative $U' = YZ' + ZY'$ also belongs to $\mathcal{D}_X^{2\alpha}$

stability of composition

Theorem (7.5. (Stability of composition) in Textbook)

Let $\mathbf{X} = (X, \mathbb{X})$, $\tilde{\mathbf{X}} = (\tilde{X}, \tilde{\mathbb{X}}) \in \mathcal{C}^\alpha$, $(Y, Y') \in \mathcal{D}_X^{2\alpha}$, $(\tilde{Y}, \tilde{Y}') \in \mathcal{D}_{\tilde{X}}^{2\alpha}$.

For $\varphi \in \mathcal{C}_b^3$ define

$$(Z, Z') := (\varphi(Y), D\varphi(Y)Y') \in \mathcal{D}_X^{2\alpha}$$

and $(\tilde{Z}, \tilde{Z}')$ similarly. Then, one has the local Lipschitz estimates

$$\begin{aligned} & d_{X, \tilde{X}, 2\alpha}(Z, Z'; \tilde{Z}, \tilde{Z}') \\ & \leq C_M((\varrho_\alpha(\mathbf{X}, \tilde{\mathbf{X}})) + |Y_0 - \tilde{Y}_0| + |Y'_0 - \tilde{Y}'_0| + d_{X, \tilde{X}, 2\alpha}(Y, Y'; \tilde{Y}, \tilde{Y}')) \end{aligned}$$

as well as

$$\begin{aligned} \|Z - \tilde{Z}\|_\alpha & \leq C_M(\varrho_\alpha(\mathbf{X}, \tilde{\mathbf{X}}) + |Y_0 - \tilde{Y}_0| + |Y'_0 - \tilde{Y}'_0| \\ & \quad + d_{X, \tilde{X}, 2\alpha}(Y, Y'; \tilde{Y}, \tilde{Y}')) \end{aligned}$$

for a constant $C_M = C(M, T, \alpha, \phi)$

Proof. Similar to the previous proof, note that

$$|Z'_0 - \tilde{Z}'_0| = |D\varphi(Y_0)Y'_0 - D\varphi(\tilde{Y}_0\tilde{Y}'_0)| \leq C_M(|Y_0 - \tilde{Y}_0| + |Y'_0 - \tilde{Y}'_0|)$$

the last inequality in the statement is an immediate consequence of the first one and the following

$$\|Y - \tilde{Y}\|_\alpha \leq C(\|X - \tilde{X}\|_\alpha + |Y'_0 - \tilde{Y}'_0| + d_{X,\tilde{X},2\alpha}(Y, Y'; \tilde{Y}, \tilde{Y}'))$$

We need to bound

$$\|D\phi(Y)Y' - D\phi(\tilde{Y})\tilde{Y}'\|_\alpha + \|R^Z - R^{\tilde{Z}}\|_{2\alpha}$$

Write $C_M(\epsilon_X + \epsilon_0 + \epsilon_{;0} + \epsilon)$ for the RHS of the first. From above

$$\|Y - \tilde{Y}\|_\alpha \lesssim \epsilon_X + \epsilon'_0 + \epsilon =: \epsilon_Y$$

and also $\|Y - \tilde{Y}\|_{\infty;[0,T]} \lesssim \epsilon_0 \epsilon_Y$ (uniformly over $T \leq 1$).

stability of composition

Since $D\varphi \in \mathcal{C}_b^2$, from Lemma 8.2

$$\begin{aligned}\|D\phi(\tilde{Y}) - D\phi(Y)\|_{\mathcal{C}^\alpha} &= |D\phi(\tilde{Y}_0) - D\phi(Y_0)| + \|D\phi(\tilde{Y}) - D\phi(Y)\|_\alpha \\ &\leq C(\epsilon_0 + \epsilon_Y)\end{aligned}$$

where C depends on the $\mathcal{C}_b^3$ -norm of ϕ . Also, $\|Y' - \tilde{Y}'\|_{\mathcal{C}^\alpha} \leq \epsilon'_0 + \epsilon$.

Clearly then ($\mathcal{C}^\alpha$ is a Banach algebra under pointwise multiplication), for a constant C_M ,

$$\|D\varphi(Y)Y' - D\varphi(\tilde{Y})\tilde{Y}'\|_\alpha \leq C_M(\epsilon_0 + \epsilon_Y + \epsilon'_0 + \epsilon) \lesssim C_M(\epsilon_X + \epsilon_0 + \epsilon'_0 + \epsilon).$$

To deal with $R^Z - R^{\tilde{Z}}$, write

$$\begin{aligned}R_{s,t}^Z &= \varphi(Y_t) - \varphi(Y_s) - D\varphi(Y_s)Y'_s X_{s,t} \\ &= \varphi(Y_t) - \varphi(Y_s) - D\varphi(Y_s)Y_{s,t} + D\varphi(Y_s)R_{s,t}^Y.\end{aligned}$$

stability of composition

Taking the difference with $R^{\tilde{Z}}$ (replace Y, Y', R^Y above by $\tilde{Y}, \tilde{Y}', R^{\tilde{Y}}$) leads to the bound $|R_{s,t}^Z - R_{s,t}^{\tilde{Z}}| \leq T_1 + T_2$ where

$$\begin{aligned} T_1 &:= \varphi(Y_t) - \varphi(Y_s) - D\varphi(Y_s)Y_{s,t} - (\varphi(\tilde{Y}_t) - \varphi(\tilde{Y}_s) - D\varphi(\tilde{Y}_s)\tilde{Y}_{s,t}) \\ &= \int_0^1 \left(D^2\varphi(Y_s + \theta Y_{s,t})(Y_{s,t}, Y_{s,t}) - D^2\varphi(\tilde{Y}_s + \theta \tilde{Y}_{s,t})(\tilde{Y}_{s,t}, \tilde{Y}_{s,t}) \right) \\ &\quad \times (1 - \theta) d\theta \end{aligned}$$

$$T_2 := D\varphi(Y_s)R_{s,t}^Y - D\varphi(\tilde{Y}_s)R_{s,t}^{\tilde{Y}}.$$

For the second term,

we know $R_{s,t}^Y - R_{s,t}^{\tilde{Y}} \leq (\epsilon'_0 + \epsilon)|t - s|^{2\alpha}$, for all s, t while

$$|D\varphi(\tilde{Y}_s) - D\varphi(Y_s)| \leq \|D^2\varphi\|_\infty |\tilde{Y}_s - Y_s| \leq \|D^2\varphi\|_\infty (\epsilon_0 + \epsilon_Y).$$

By elementary estimates of the form $|ab - \tilde{a}\tilde{b}| \leq |a||b - \tilde{b}| + |a - \tilde{a}||\tilde{b}|$ it then follows immediately that one has

$$T_2 \leq C(\epsilon_X + \epsilon_0 + \epsilon'_0 + \epsilon)|t - s|^{2\alpha}.$$

For the first term,

Consider the expression under the integral $\int(\dots)(1-\theta)d\theta$ for fixed integration variable $\theta \in [0, 1]$. Using $Y^n \rightarrow Y$ in α -Hölder norm,

$$\begin{aligned} & |D^2\varphi(\tilde{Y}_s + \theta\tilde{Y}_{s,t}) - D^2\varphi(Y_s + \theta Y_{s,t})| \\ & \leq \|D^3\varphi\|_\infty (|\tilde{Y}_s - Y_s| + |\tilde{Y}_{s,t} - Y_{s,t}|) \\ & \leq 3\|D^3\varphi\|_\infty \lesssim \epsilon_0 + \epsilon_Y, \end{aligned}$$

noting that this estimate is uniform in $s, t \in [0, T]$ and $\theta \in [0, 1]$.

It then suffices to insert/subtract $D^2\varphi(Y_s + \theta Y_{s,t})(\tilde{Y}_{s,t}, \tilde{Y}_{s,t})$ under the integral $\int \dots (1-\theta)d\theta$ appearing in the definition of T_1 and conclude with the triangle inequality and some simple estimates, keeping in mind that $\|Y - \tilde{Y}\|_\alpha \leq \epsilon_Y$ and $\|Y\|_\alpha, \|\tilde{Y}\|_\alpha \lesssim C_M$. $\square$

Itô's formula revisited

Proposition (5.6. Itô formula for reduced rough paths in Textbook)

Let $F : V \rightarrow W$ be of class $\mathcal{C}_b^3$ and let $\mathbf{X} = (X, \mathbb{S}) \in \mathcal{C}_r^\alpha([0, T], V)$ with $\alpha > 1/3$. Then

$$F(X_t) - F(X_0) = \int_0^t DF(X_s) d\mathbf{X}_s + \frac{1}{2} \int_0^t D^2 F(X_s) d[\mathbf{X}]_s, \quad 0 \leq t \leq T.$$

Proposition (7.6 in Textbook)

Let $F : V \rightarrow W$ in $\mathcal{C}_b^3$, $\mathbf{X} = (X, \mathbf{X}) \in \mathcal{C}^\alpha$ and $(Y, Y') \in \mathcal{D}_X^{2\alpha}$ a controlled rough path of the form

$$Y_t = \int_0^t Y'_s d\mathbf{X}_s + \Gamma_t$$

for some controlled rough path $(Y', Y'') \in \mathcal{D}_X^{2\alpha}$ and some path $\Gamma \in \mathcal{C}^{2\alpha}$. The increments of Y are of the form

$$Y_{s,t} = Y'_s X_{s,t} + Y''_s \mathbb{X}_{s,t} + \Gamma_{s,t} + o(|t - s|).$$

Then the following Itô formula holds.

$$\begin{aligned} F(Y_t) &= F(Y_0) + \int_0^t DF(Y_s) Y'_s d\mathbf{X}_s + \int_0^t DF(Y_s) d\Gamma_s \\ &\quad + \frac{1}{2} \int_0^t D^2 F(Y_s) (Y'_s, Y'_s) d[\mathbf{X}]_s. \end{aligned}$$

Proof. By the (previous) Itô formula,

$$F(Y_t) - F(Y_0) = \lim_{|\mathcal{D}| \rightarrow 0} \sum_{[u,v] \in \mathcal{D}} DF(Y_u)Y_{u,v} + D^2F(Y_u)\mathbb{Y}_{u,v} + \lim_{|\mathcal{D}| \rightarrow 0} \sum_{[u,v] \in \mathcal{D}} D^2F(Y_u)[\mathbf{Y}]_{u,v}$$

where $\mathbb{Y}_{u,v} = \int_u^v Y_{u,\cdot} \otimes dY$ in the sense of remark 4.11., noting that $\mathbb{Y}_{u,v} = Y'_u Y'_u \mathbb{X}_{u,v} + o(|v - u|)$. Also,

$$\begin{aligned} [\mathbf{Y}_{u,v}] &= Y_{u,v} \otimes Y_{u,v} - 2\text{Sym}(\mathbb{Y})_{u,v} \\ &= Y'_u Y'_u (X_{u,v} \otimes X_{u,v} - 2\text{Sym}(\mathbb{X}_{u,v})) + o(|v - u|) \\ &= Y'_u Y'_u [\mathbf{X}]_{u,v} + o(|v - u|). \end{aligned}$$

On the other hands, by lemma 7.3 4, the path $G := DF(Y)Y'$ is controlled by X , with $G' = D^2F(Y)(Y', Y') + DF(Y)Y''$, so that

$$\begin{aligned}\int DF(Y_s)Y'_s d\mathbf{X}_s &= \int_0^t G_s d\mathbf{X}_s \\&= \lim_{|P| \rightarrow 0} \sum_{[u,v] \in P} G_u X_{u,v} + G'_u \mathbb{X}_{u,v} \\&= \lim_{\mathcal{D} \rightarrow 0} \sum_{[u,v] \in \mathcal{D}} (DF(Y_u)Y'_u X_{u,v} + (DF(Y_u)Y''_u + D^2F(Y_u)Y'_u Y'_u) \mathbb{X}_{u,v})\end{aligned}$$

Itô's formula revisited

Let us subtract/add $DF(Y_u)Y_u''\mathbb{X}_{u,v}$ from the first equality in this proof to complete.

$$\begin{aligned}
 & F(Y_t) - F(Y_0) \\
 &= \lim_{|\mathcal{D}| \rightarrow 0} \sum_{[u,v] \in \mathcal{D} \rightarrow 0} (DF(Y_u)(Y_{u,v} - Y_u''\mathbb{X}_{u,v}) + DF(Y_u)Y_u''\mathbb{X}_{u,v} \\
 &\quad + D^2F(Y_u)Y_u'Y_u'\mathbb{X}_{u,v}) + \lim_{\mathcal{D} \rightarrow 0} \sum_{[u,v] \in \mathcal{D}} D^2F(Y_u)Y_u'Y_u'[\mathbf{X}]_{u,v} \\
 &= \lim_{\mathcal{D} \rightarrow 0} \sum_{[u,v] \in \mathcal{D}} (DF(Y_u)Y_u'X_{u,v} + (DF(Y_u)Y_u'' + D^2F(Y_u)Y_u'Y_u')\mathbb{X}_{u,v}) \\
 &\quad + \lim_{\mathcal{D} \rightarrow 0} \sum_{[u,v] \in \mathcal{D}} DF(Y_u)\Gamma_{u,v} + \int_0^t D^2F(Y_u)Y_u'Y_u'd[\mathbf{X}]_u. \\
 &= \int_0^t DF(Y_s)Y_s'd\mathbf{X}_s + \int_0^t DF(Y_s)d\Gamma_s + \frac{1}{2} \int_0^t D^2F(Y_s)(Y_s', Y_s')d[\mathbf{X}]_s.
 \end{aligned}$$

□

Controlled rough paths of low regularity

Controlled rough paths of low regularity

Recall that we view a controlled rough path Y as a $T^{(p-1)}(\mathbf{R}^d)$ -valued function.

For any alphabet $\mathcal{A}$, the shuffle product $\sqcup$ is defined on the free algebra over $\mathcal{A}$, by considering all possible ways of interleaving two words in ways that preserve the original order of the letter. For example,

$$ab \sqcup ac = abac + 2aabc + 2aacb + acab (a, b, c \in \mathcal{A})$$

In our case, the choice of the basis described earlier then defines a natural algebra homomorphism $\omega \mapsto e_\omega$ from the free algebra over $\{1, \dots, d\}$ into $T^{(p)}(\mathbf{R})^d$, and we denote by $\star$ the corresponding product.

$$e_\omega \star e_{\bar{\omega}} = e_{\omega \sqcup \bar{\omega}}$$

where, if ω is a linear combination of words, e_ω is the corresponding linear combination of basis vectors.

Controlled rough paths of low regularity

Any geometric rough path $\mathbf{X}$ satisfies

$$\mathbf{X}_{s,t}^\omega \mathbf{X}_{s,t}^{\bar{\omega}} = \mathbf{X}_{s,t}^{\omega \sqcup \bar{\omega}}.$$

The way multiplying two controlled rough paths Y and $\bar{Y}$ is to define their product Z by

$$Z_t = Y_t \star \bar{Y}_t.$$

Z is a controlled rough path. Similarly, if F is a smooth function and Y is a controlled rough path, we define $F(Y)$ by

$$F(Y)_t \stackrel{\text{def}}{=} F(Y_t^\phi) + \sum_{k=1}^{p-1} \frac{1}{k!} F^{(k)}(Y_t^\phi) \tilde{Y}_t^{\star k},$$

where $F^{(k)}$ denotes the k th derivative of F and $\tilde{Y}_t \stackrel{\text{def}}{=} Y_t - Y_t^\phi$ is the part describing the "local fluctuations" of Y .

Summary

1. There is a continuous canonical injection

$$\mathcal{D}_X^{2\alpha}([0, T], W) \hookrightarrow \mathcal{C}^\alpha([0, T], W)$$

2. The chain rule holds for a composition with regular functions.
3. For a composition with regular functions, one has the local Lipschitz estimates

$$\begin{aligned} & d_{X, \tilde{X}, 2\alpha}(Z, Z'; \tilde{Z}, \tilde{Z}') \\ & \leq C_M((\varrho_\alpha(\mathbf{X}, \tilde{\mathbf{X}})) + |Y_0 - \tilde{Y}_0| + |Y'_0 - \tilde{Y}'_0| + d_{X, \tilde{X}, 2\alpha}(Y, Y'; \tilde{Y}, \tilde{Y}')) \end{aligned}$$

4. Itô formula holds for a controlled rough path.
5. One may define the product of two controlled rough paths so that the product is again a controlled rough path.

THANK YOU FOR LISTENING